Polish-speaking children show facilitative use of gender marking for both high-frequency and low-frequency adjectives.

Background. Adults recognize nouns faster when preceded by informative grammatical gender cues, an effect documented across languages **[1,2]**. While most research has focused on article-marking systems, similar facilitation has been observed in languages that mark gender on adjectives, including Polish **[1,3]**. Developmental evidence remains limited, particularly in article-less systems. Although children can use prenominal adjectival information to anticipate nouns, less is known about how this ability develops in Polish and how it is shaped by lexical frequency. Usage-based models propose that linguistic representations emerge gradually and are influenced by frequency at multiple levels **[4,5].** We therefore tested whether Polish-speaking 7-year-olds use prenominal gender marking on adjectives to anticipate upcoming nouns, and whether adjective frequency modulates this process. Frequency-independent effects would suggest reliance on abstract morphological cues, whereas effects restricted to high-frequency adjectives would suggest that children at that age still rely on whole lexical items.
Aims. This study tests whether Polish-speaking children use gender-marked adjectives to anticipate nouns, and whether adjective frequency modulates this effect, as predicted by usage-based models.
Methods. Polish-speaking 7-year-olds (n = 31) completed a Visual World eye-tracking task. Each trial presented two same-colored images (green or red) alongside an auditory prompt **[Example 1]**. Eight target nouns (4 masculine, 4 feminine) were preceded by a gender-marked color adjective—either high-frequency (*zielon-y/a* 'green', *czerwon-y/a* 'red') or low-frequency (*szałwiow-y/a* 'sage', *szkarłatn-y/a* 'scarlet')—based on CDS frequencies **[6].** The adjective was always inflected to match the gender of the target noun. The distractor shared or differed in grammatical gender from the target **[Figure 1].** 32 trials were counterbalanced across two lists so that each target appeared with only one adjective form. Prior to the experiment, children were familiarized with the pictures and were asked to name them to ensure knowledge of the corresponding labels.
Predictions. Given prior work, we expect to find evidence of facilitative processing of gender, evidenced by a higher proportion of fixations to the target noun in "different" as compared to the "same" condition trials. A usage-based perspective predicts that this effect to be smaller for low-frequency adjectives, reflecting reduced exposure to specific (inflected) forms.
Results. A cluster-based permutation analysis **[7]** was used to identify time bins in which looks to target differed by condition and/or frequency **[Figure 2]**. The analysis found a significant cluster for condition in the expected direction (different > same) at 4350-4900ms (cluster mass statistic = 120.0, $p<0.001$), as well as a significant cluster in the opposite direction at 5200-5500ms (cluster mass statistic = 62.3, $p<0.001$). However, no significant cluster was found for the effect of frequency, or for the interaction between frequency and condition. An additional post-hoc divergence point analysis **[Figure 3]** compared the divergence point for the different conditions when the adjective was frequent (4697ms; CI: (4600,4800)) to when the adjective was non-frequent (4566ms; CI: (4500,4700)) and found no significant difference ($p=0.80$).
Implications. The results show that Polish-speaking children can use gender marking on adjectives to anticipate upcoming nouns. This supports the role of grammatical gender as a predictive cue in real-time processing by early mid-childhood. While anticipatory effects were found for both frequent and non-frequent adjectives, no evidence was found for an effect of adjective frequency on the timing or strength of this process.

Example 1. *Sample auditory prompt*

Pokaż,	gdzie	jest	zielon-y	guzik?
indicate.IMP	where	be.3SG.PRS	green-M.SG.NOM	button$_M$.SG.NOM

"Show, where is (the) green button?"

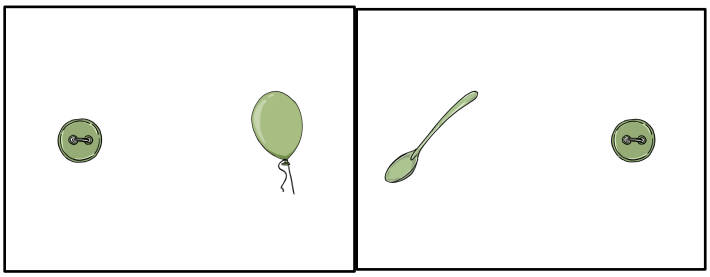


Figure 1. (a) Sample display for a same condition trial with a masculine target(*guzik* 'button$_M$'; *balon* 'balloon$_M$'); (b) sample display for a different condition trial with a masculine target (*guzik* 'button$_M$'; *łyżka* 'spoon$_F$').

Figure 2. Proportion of fixations to target, grouped by condition and by adjective frequency.

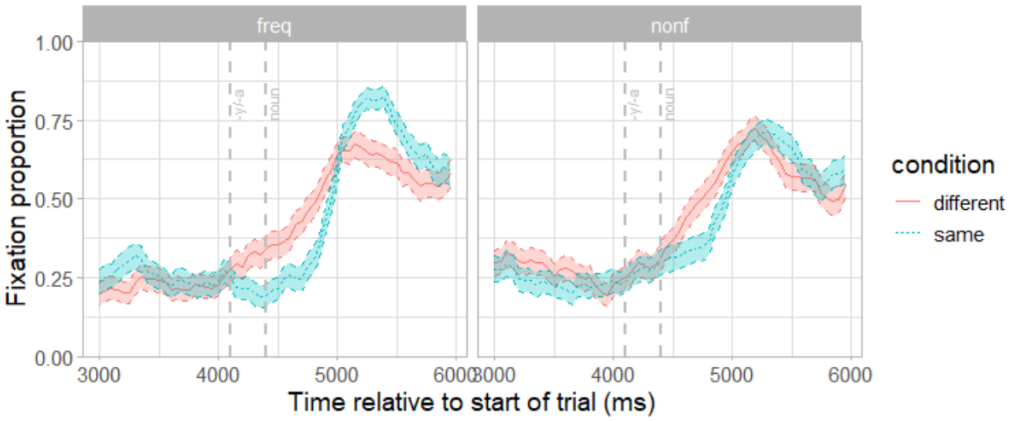
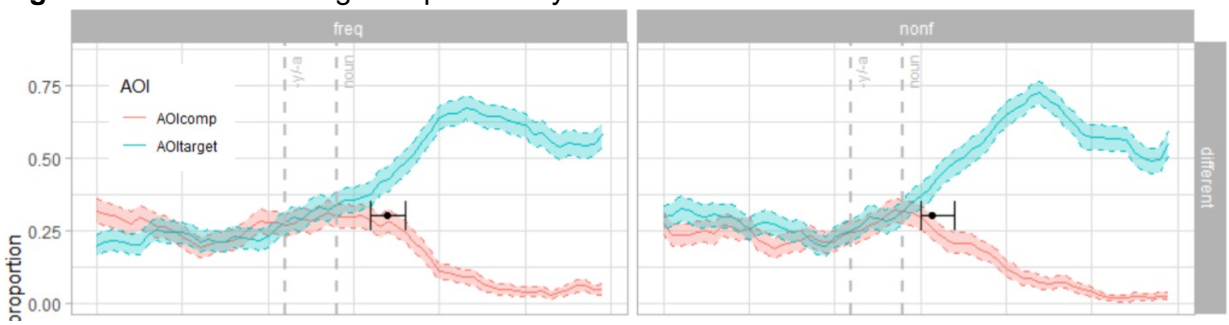


Figure 3. Results of divergence point analyses for *different* conditions.



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